

EE 252: EMPEL - *DC-DC Buck-Boost Converter*

Final Report: *B2-G6*

Problem Statement:

- Design and hardware implementation of a *non-isolated DC-DC Buck-Boost Converter*.
Specifications: Input 15 V, Output 8-18 V, Switching frequency 12.5 kHz, Output current 1 A.
- To record waveforms of switch current and diode voltage in CCM and increase load resistance to demonstrate DCM.

Batch & Group No.: B2-G6

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Hardware Implementation

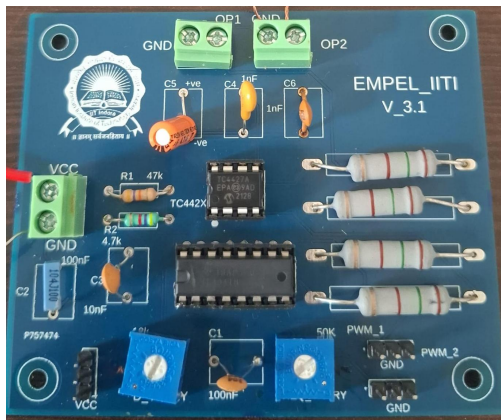


Figure 1: Soldered TL494 Gate Driver Board.

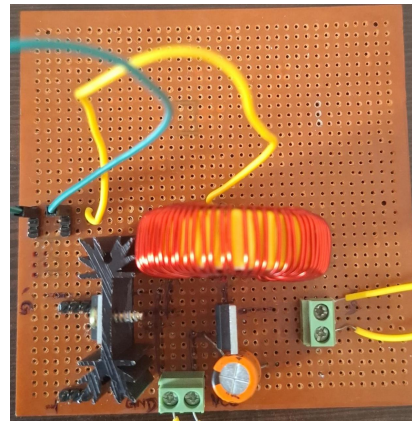


Figure 2: Buck-Boost Converter Power Circuit.

Experimental Results

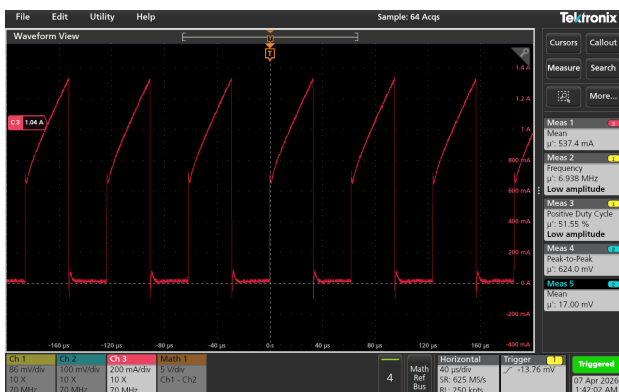


Figure 3: Switch current in CCM

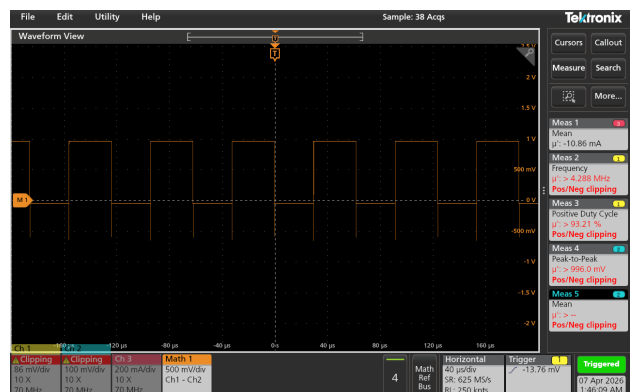


Figure 4: Diode Voltage in CCM



Figure 5: Inductor current in CCM ($R = 5.6\Omega$)
Note: Continuous inductor current above 200mA.

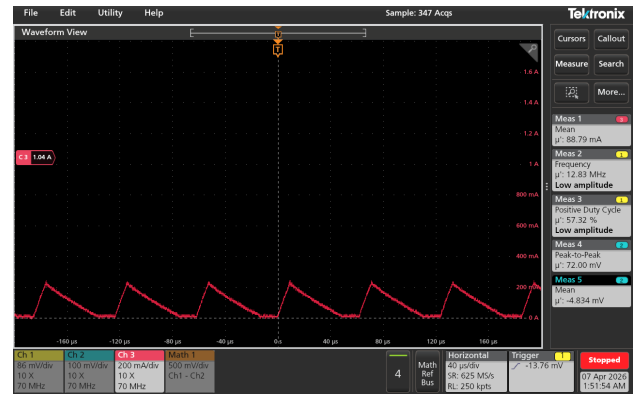


Figure 6: Inductor current in DCM ($R = 36.8\Omega$)
Note: Discontinuous current with increased load.

Simulated waveforms

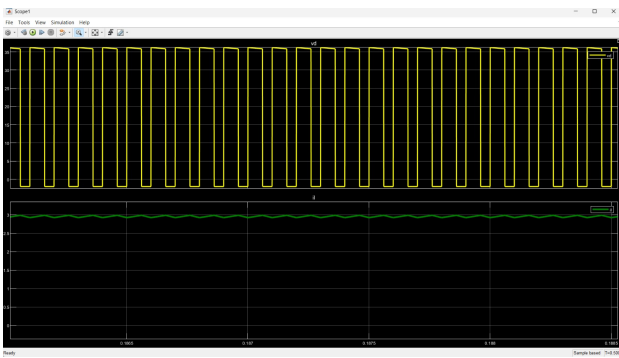


Figure 7: Diode voltage and inductor current

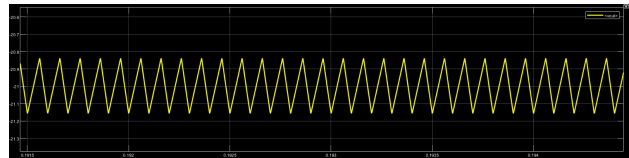


Figure 8: Output Voltage

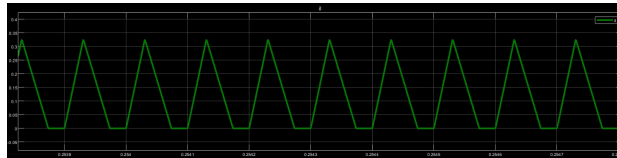


Figure 9: Inductor current at DCM

Demo Video

[Drive Link for the video](#)

Testing

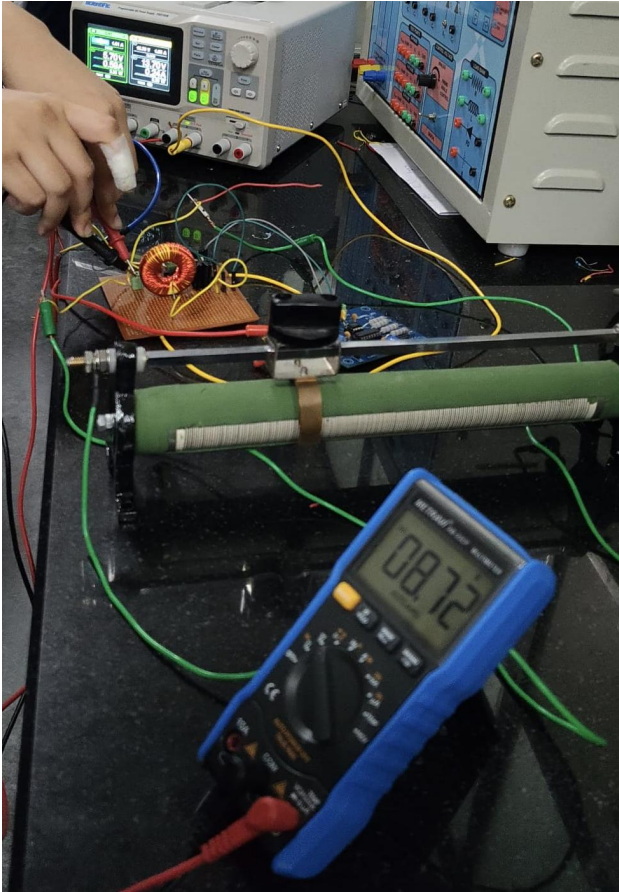


Figure 10:
Input voltage = 5.70V, Output measured = 8.72V,
at $D = 67\%$

Conclusion: Boost mode tested

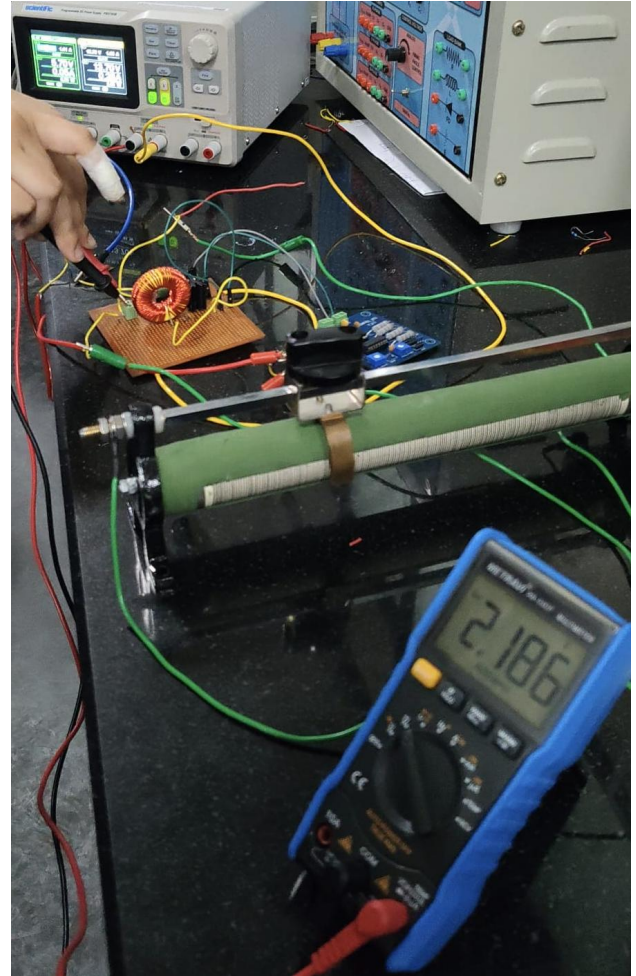


Figure 11:
Input voltage = 5.70V, Output measured = 2.186V,
at $D = 35\%$

Conclusion: Buck mode tested